Kishoreganj University

3rd Year 1st Semester B. Sc. (Engg.) First In-course Examination, 2024 (Session: 2021-22)

Department of Computer Science and Engineering

CSE 3105: Compiler Design (3 Credits)

Time: 1 Hour **Marks: 25**

Answer all the questions

[The figures in the right margin indicate the marks. All portions of each question must be answered sequentially.]

1. a. Describe different types of errors with proper examples. 3

   b. Find out if the following CFG is ambiguous or not, and give proper justification. You may use "**abaabbababababaa**" as input string to test the ambiguousness. 4

   S-> A B A | aaBA
   A-> aBa | bA
   B-> bBB | ba

2. In physics, equations of motions are equations that describe the behavior of a physical system in terms of its motion as a function of time. One of the equations of motion can be described by the following equation: 8

   *s=u*t+(1/2)*a*t*t*

   Consider that you have written a code containing the equation given above. Determine the output of each phase of the compiler for the equation. Consider that the instructions for loading, multiplying, adding, dividing, and storing are LDF, MULF, ADDF, DIVF, and STF, respectively in the final code generation step.

3. a. Consider the following grammar and perform left factoring: 4

   P→ rPPssP | rPPsPr | sPs | r | rX
   X→ pqX | pXq | pqrX | p

   b. Write down the regular expressions for the following languages: 6

   I. Consider **Σ = {a, b, c, d}**. Identify the regular expression for strings that start with an odd number of **a**'s or an even number of **b**'s, followed by exactly three **c**'s. Additionally, the second letter from the right in the string is always **d**.

   II. A binary number that starts with either 0 or 1 and is always an even number.